Distributionally Consistent Simulation of Naturalistic Driving Environment for Autonomous Vehicle Testing

Xintao Yan, *Member, IEEE*, Shuo Feng, *Member, IEEE*, Haowei Sun, and Henry X. Liu, *Senior Member, IEEE*

***Abstract*— Microscopic traffic simulation provides a controllable, repeatable, and efficient testing environment for autonomous vehicles (AVs). To evaluate AVs' safety performance unbiasedly, the probability distributions of environment statistics in the simulated naturalistic driving environment (NDE) need to be consistent with those from the real-world driving environment. However, although human driving behaviors have been extensively investigated in the transportation engineering field, most existing models were developed for traffic flow analysis without considering the distributional consistency of driving behaviors, which could cause significant evaluation biasedness for AV testing. To fill this research gap, a distributionally consistent NDE modeling framework is proposed in this paper. Using large-scale naturalistic driving data, empirical distributions are obtained to construct the stochastic human driving behavior models under different conditions. To address the error accumulation problem during the simulation, an optimization-based method is further designed to refine the empirical behavior models. Specifically, the vehicle state evolution is modeled as a Markov chain and its stationary distribution is twisted to match the distribution from the real-world driving environment. The framework is evaluated in the case study of a multi-lane highway driving simulation, where the distributional accuracy of the generated NDE is validated and the safety performance of an AV model is effectively evaluated.**

***Index Terms*— Autonomous vehicle, driving behavior, naturalistic driving environment, simulation, testing and evaluation.**

I. Introduction

TESTING and evaluation is a critical step in the development and deployment of autonomous vehicles (AVs), which has received extensive attention from both the industry and academia in recent years [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16]. Prevailing methods test AVs in the naturalistic driving environment (NDE), observe their performance, and make statistical comparisons to human driving performance [13]. Due to the rareness of safety-critical events, however, it has been pointed out that hundreds of millions and sometimes hundreds of billions of miles would be required in NDE to demonstrate the safety performance of AVs at the level of human-driven vehicles, which is intolerably inefficient for the real-world testing [1]. Therefore, testing AVs in NDE simulations has attracted increasing attention because of the advantages of controllability, repeatability, and efficiency [13], [17].

The key to simulation testing is the trustworthiness of the testing results. As pointed out in various domains [17], [18], [19], [20], the sim-to-real gap could hinder and even mislead the training and testing process of an agent. To fill this gap, existing studies have paid much attention to the fidelity of vehicle dynamics, sensor models, and photorealistic images based on techniques such as computer graphics, physics-based modeling, and data augmentation [17], [21], [22]. However, how to model the naturalistic behavior of human-driven vehicles with high fidelity still remains an open question. To answer this question, many AV companies tried to replay human driving behaviors according to the logged data collected from the real-world driving environment. However, as the human driving behaviors are pre-determined in the logged data, they cannot interact with AV models, which severely limits the scenarios that can be simulated. To address this issue, the human driving models developed in the transportation engineering field have been applied, such as the Intelligent Driving Model (IDM) [23] and MOBIL [24] models. However, although these models can interact with AV models, they were designed for traffic flow analysis purposes such as reproducing traffic oscillations and the fundamental diagram, which are not suitable for the AV simulation testing (see Section II for more details of related studies).

We first use an illustration example to demonstrate the importance of simulation accuracy. Consider a car-following scenario shown in the left side of Fig. 1. We assume that the distance between the two vehicles in the real-world driving environment one step later follows the red-curve distribution, as shown in the middle of Fig. 1. If the simulation is inaccurate, the simulated distance between the two vehicles

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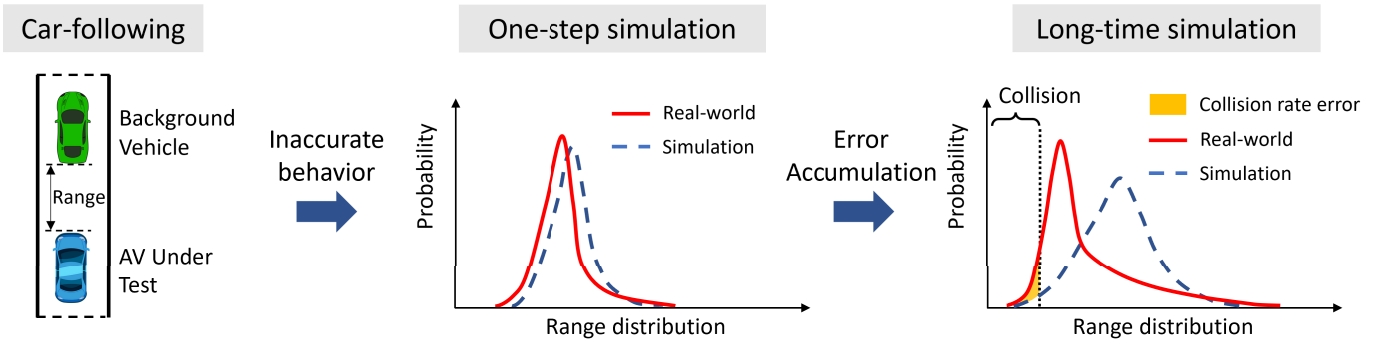


Fig. 1. Illustration of the distribution inconsistency that might mislead AV development and deployment.

will follow a different distribution, say, the dashed blue curve in the middle of Fig. 1. This distributional discrepancy will be compounded and amplified during the simulation process, which could result in a significant difference, as shown in the right of Fig. 1. This inconsistency can cause sim-to-real gaps, such as a lower simulated crash rate than the real-world crash rate. If the following vehicle is an AV, this optimistic estimation of the AV safety performance might be misleading and result in misguided AV development.

In this paragraph, we will mathematically demonstrate the new NDE modeling requirement brought by AV testing applications. To evaluate AVs' safety performance quantitatively, the accident rate of AVs in NDE is usually utilized [2], [7], [8], [13]. As the human driving behaviors significantly affect the response and performance of AVs, estimating the accident rate accurately requires the NDE model in simulation to be distributionally consistent with the real-world driving environment. Specifically, let X denote the variables that define the environment, such as the behaviors of human drivers. Then, testing an AV in NDE is essential to sample X from its underlying distribution, denoted as $X \sim P(X)$, to estimate its performance μ_E^κ by

$$\mu_E^\kappa := \mathbb{E}_X\left(\phi_E^\kappa(X)\right) \approx \frac{1}{n}\sum_{i=1}^{n}\phi_E^\kappa(X_i) \approx \frac{m}{n}, X_i \sim P(X), \tag{1}$$

where E denotes the event of interest (e.g., crash), κ denotes the AV agent under test, $\phi_E^\kappa(X)$ denotes the AV performance at the environment specified by X, n denotes the number of tests, and m denotes the number of event E occurred during tests. This equation represents the crash rate calculation of an AV using the Monte Carlo simulation method, i.e., let the AV run in the NDE simulation and the crash rate is calculated by dividing the number of crashes by the total number of test runs. The Monte Carlo method [25] requires a statistically accurate simulation environment to obtain an unbiased estimation result. Therefore, the AV simulation testing requires distributionally consistent NDE models, which are significantly different from those for traffic flow analysis. To the best of our knowledge, we are the first to identify and systematically investigate this new NDE modeling requirement, and there is no existing method that can fulfill this requirement.

To fill this research gap, we propose a data-driven optimization-based NDE modeling framework in this paper, which can ensure the distributional consistency of vehicle microscopic behaviors. The overall pipeline of the framework includes two major steps as shown in Fig. 2. In the first data-driven step, we propose to directly construct human driving behavior models using empirical distributions from the large-scale naturalistic driving data (NDD). Specifically, the driving behaviors are modeled by two longitudinal models (i.e., free driving and car following) and four lateral models considering different driving situations, and each model is described as action distributions for different states. For example, the car-following behaviors can be modeled by the acceleration distribution of the following vehicle conditional on different self speeds, relative speeds, and relative distances with the preceding vehicle. These empirical behavior models serve as basic models. If the dataset is sufficiently accurate, diverse, and large, the empirical distributions can accurately characterize human driving behaviors in different situations. However, due to the limitation of data quantity and unavoidable data noise, the obtained empirical distributions may still have small inaccuracy compared with the ground truth. As illustrated in Fig. 1, the small inaccuracy can be accumulated and compounded with the simulation, so the resultant simulation environment will deviate from the realistic environment.

To further tackle the error accumulation issue, the second step of the framework is to refine empirical behavior models to minimize the accumulated errors using optimization methods. To achieve this goal, the key is to model the long-term effects of the error accumulation. In this study, by modeling the vehicle state evolution as a Markov chain, the long-term effects of the error accumulation could be modeled based on the stationary distribution of the Markov chain. Then, an optimization problem can be formulated to minimize the accumulated errors by adjusting the empirical behavior models. In this way, the error accumulation could be reduced, which results in the NDE model with more accurate distributions. This provides opportunities to conduct high-fidelity long-time simulation for full-length trip evaluation of AV's performance [26]. Using the large-scale real-world NDD collected by the University of Michigan, Ann Arbor, we validate the performance of the proposed method for

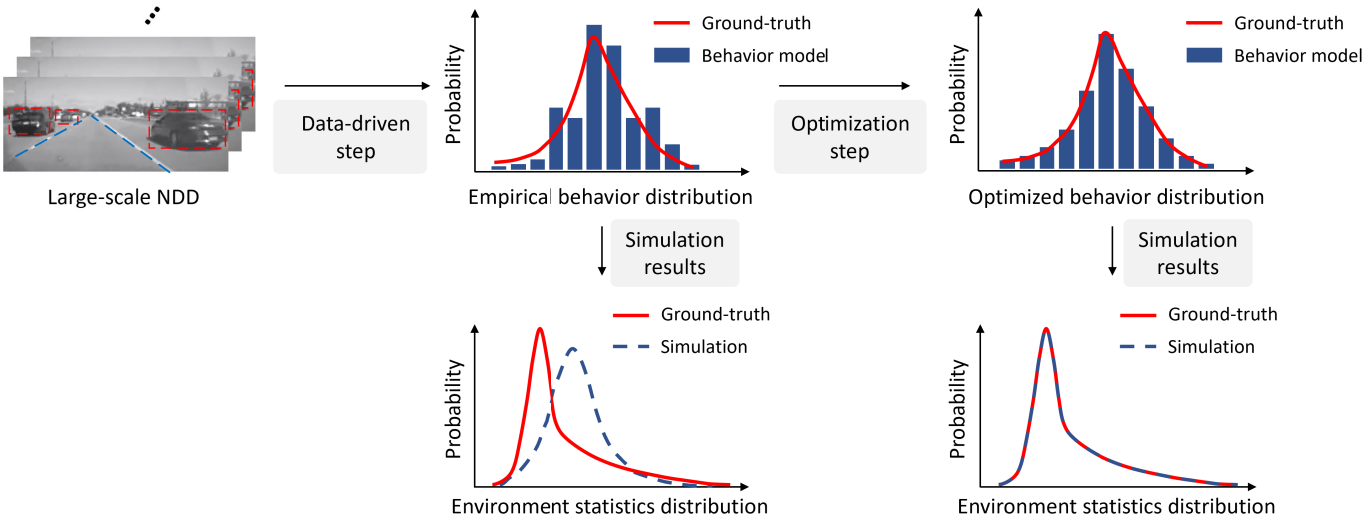


Fig. 2. Proposed framework pipeline.

a multilane highway driving environment. Compared with existing models, the proposed method demonstrates superior performance regarding the distributional accuracy. To further validate the capability for AV testing, the generated NDE is also utilized to test the safety performance of an AV agent.

In summary, the main contributions of this paper are threefold: first, the new requirements of NDE modeling are identified for AV testing purposes, which cannot be satisfied by most existing methods; second, a novel modeling framework is proposed to generate the NDE that is distributionally consistent with the real-world driving environment; third, the proposed method is validated using large-scale real-world NDD and the generated NDE is further validated by testing an AV model.

The rest of this paper is organized as follows. In Section II, we provide a brief review of existing NDE modeling methods. In Section III, we introduce the data-driven step to construct empirical behavior models using large-scale real-world NDD and evaluate their performances in a multilane highway driving environment simulation. In Section IV, to account for the error accumulation problem, the robust modeling step is proposed to optimize the empirical behavior models to obtain a distributionally consistent NDE. In Section V, the performance of the proposed method is validated and compared with existing models. An AV testing experiment is further conducted to demonstrate the capability of the proposed NDE in this section. Finally, Section VI concludes the paper and lays out some future directions.

II. Related Work

In this section, we provide a brief overview of existing driving behavior models, especially focusing on how the behavior stochasticity is modeled and whether they can achieve distribution-level accuracy. A general form of the existing driving behavior model can be expressed by

$$u(t) = \psi(S(t), \theta(t)) + \epsilon(t), \tag{2}$$

where t denotes the time, $u(t)$ denotes the action (e.g., longitudinal acceleration) of the vehicle at the t-th time instance, $S(t)$ denotes the states of the ego-vehicle and surrounding vehicles that have the influence on the ego-vehicle's decision making, $\psi(\cdot)$ denotes the model that maps from the state space to the action space, $\theta(t)$ denotes model parameters that could be deterministic or stochastic, and $\epsilon(t)$ denotes the additive noise term.

Most traditional models [23], [24], [27], [28], [29] are deterministic and cannot capture the stochastic nature of human driving behaviors. For these models, the model parameters θ remain fixed during the simulation and no external noise term ϵ is added. Although the parameters of these models can be calibrated using real-world data [30], [31], [32], [33], [34], they cannot capture the stochastic behaviors of human drivers. Recently, increasing studies have used machine learning-based methods to fit the behavior model ψ using neural networks [35], [36], [37], [38], [39]. By utilizing large-scale naturalistic driving datasets, these methods aim to better reproduce the observed trajectories of human drivers. However, the problem of lacking accurate stochasticity still remains unsolved.

The stochasticity can be incorporated into the model to a certain extent by introducing external noise term $\epsilon(t)$ and/or stochasticity to model parameters $\theta(t)$. For external noise $\epsilon(t)$, the most commonly used one is the Gaussian noise [40], [41], [42], [43], [44]. However, the external addition of the Gaussian noise cannot realistically depict human driving behaviors since the interaction in different driving conditions is highly complex and does not always follow the simple Gaussian distribution [45], [46], [47]. The inaccuracy may not affect significantly for the traffic flow analysis, for example, these models can reproduce real-world traffic phenomena like traffic oscillations and the fundamental diagram. However, it will cause significant evaluation biasedness on AV performance. In a few studies [46], the log-normal and extreme value distributions are applied. Although these distributions are more

realistic than the Gaussian, the accuracy is still limited due to the model flexibility for fitting complicated driving behaviors in all situations using the same distribution.

Besides adding an external noise $\epsilon(t)$, the stochasticity can also be incorporated into model parameters $\theta(t)$. For example, Treiber and Kesting [42] proposed indifference regions in the form of action points to the model, so the new acceleration will be executed only when the change is greater than a threshold parameter, which follows a uniform distribution. Yang and Peng [46] considered different human errors during car-following situations, where the length of time delay and distraction interval are modeled by exponential distribution and lognormal distribution according to real-world data, respectively. Game theory is applied in [48] to model stochastic lane-change behavior. Hamdar et al. [49], [50] proposed a utility-based stochastic car-following model where acceleration probability follows a continuous logit model. Again, these methods did not consider the distributional accuracy of the driving behaviors after the stochasticity was introduced. Therefore, although traffic simulations have been used in many applications in literature [51], [52], [53], [54], [55], [56], [57], they are not suitable for AV testing. There were two notable exceptions in [58] and [59], the authors proposed stochastic car-following models to capture the distribution of time headway. They validate that the headway distribution of the simulation environment is consistent with the real world. However, only car-following behavior is considered in these studies, so they can only simulate the single-lane road. More importantly, the error accumulation issue has not been considered, which could severely distort the distribution for long-term simulation of NDE. More related studies can be found in [60], [61], and [62] and references therein.

Notwithstanding the related studies, most existing methods cannot capture the accurate distribution of stochastic human driving behaviors. Therefore, the driving environment that is generated by these behavior models cannot accurately reproduce simulation environment statistics, which is critical for AV testing.

III. Data-Driven NDE Modeling

In this section, we propose a simple yet effective data-driven method for NDE modeling leveraging large-scale NDD. Specifically, six empirical behavior models are constructed including free-driving, car-following, and four lane-changing behaviors with different driving conditions, and then the NDE can be generated by combining the six empirical behavior models according to the driving condition at each time step (Section III-A). To construct each empirical behavior model, the large-scale NDD is processed and utilized in Section III-B. Then, a multi-lane highway driving environment is simulated to evaluate the performance of the empirical behavior models in Section III-C, which validates the data-driven method and further motivates the robust modeling step in Section IV.

A. Empirical Behavior Models

To construct the NDE, both longitudinal and lateral behaviors of human drivers need to be modeled based on the vehicle's own state and its surrounding situations S. For example, human drivers need to decide their longitudinal accelerations and decide whether to take a lateral lane-change maneuver. In this study, six behavior models are proposed including free-driving, car-following, and four lane-changing behaviors with different driving conditions, as shown in Fig. 3a. Specifically, the vehicle acceleration in the free-driving case is modeled that depends only on its current velocity, while the acceleration in the car-following case is modeled that depends on the velocity, range (relative position), and range rate (relative speed) of the subject vehicle and its preceding vehicle. The threshold between free-driving and car-following is set at 115 meters because surrounding vehicles beyond this distance from the subject vehicle are not detected and included in the dataset. To capture the lane-changing probability in different conditions, four lane-changing models are proposed, which output the lane change probability of the subject vehicle at each moment. For example, in the cut-in lane change situation, the lane-changing probability depends on velocities and distances between the subject vehicle and the preceding vehicle in the current lane and the vehicle behind in the target lane. We note that more lane-changing models could be constructed in this framework by dividing the driving conditions into more categories if needed.

After constructing the six behavior models, the NDE can be generated by combining the behavior models according to the driving condition at each time step. Taking Fig. 3b (left) as an example, the subject vehicle can take left lane change, keep car-following, or take right lane change at the moment. The left lane change behavior can be categorized as a cut-in behavior, where the lane-changing probability P_1 can be obtained from the cut-in behavior model. The right lane change behavior can be categorized as a lane change with one adjacent vehicle, where the lane-changing probability P_2 can also be obtained by the corresponding behavior model. Moreover, the longitudinal acceleration probability P_3 can be obtained by the car-following behavior model. After normalization, we can obtain the action distribution of the vehicle as shown in Fig. 3b (right). Then, the subject vehicle's action will be sampled from this distribution and used to update its state to the next time step. To simplify the modeling process, longitudinal acceleration is assumed zero if the vehicle is making a lane change behavior. Also, if there is no vehicle in front, the ego-vehicle will not take lane-changing behavior. By repeating this process for all vehicles and time steps, the NDE can be generated.

The remaining question is how to construct the six behavior models with distributional accuracy. In this study, we propose to directly estimate empirical behavior distributions as the behavior models by leveraging large-scale NDD. As the NDD records all the information needed for the human driving behaviors, accurate empirical behavior models could be constructed if using sufficient amount of data with perfect quality. Although the actual data is usually limited by the data quality and quantity, these empirical behavior models could provide a good foundation and can be further improved as discussed in Section IV. Specifically, for each behavior model, we obtain the empirical probability $P(a|S)$ for all the vehicle actions $a \in \mathcal{A}$ at all discretized states $S \in \mathcal{S}$, where

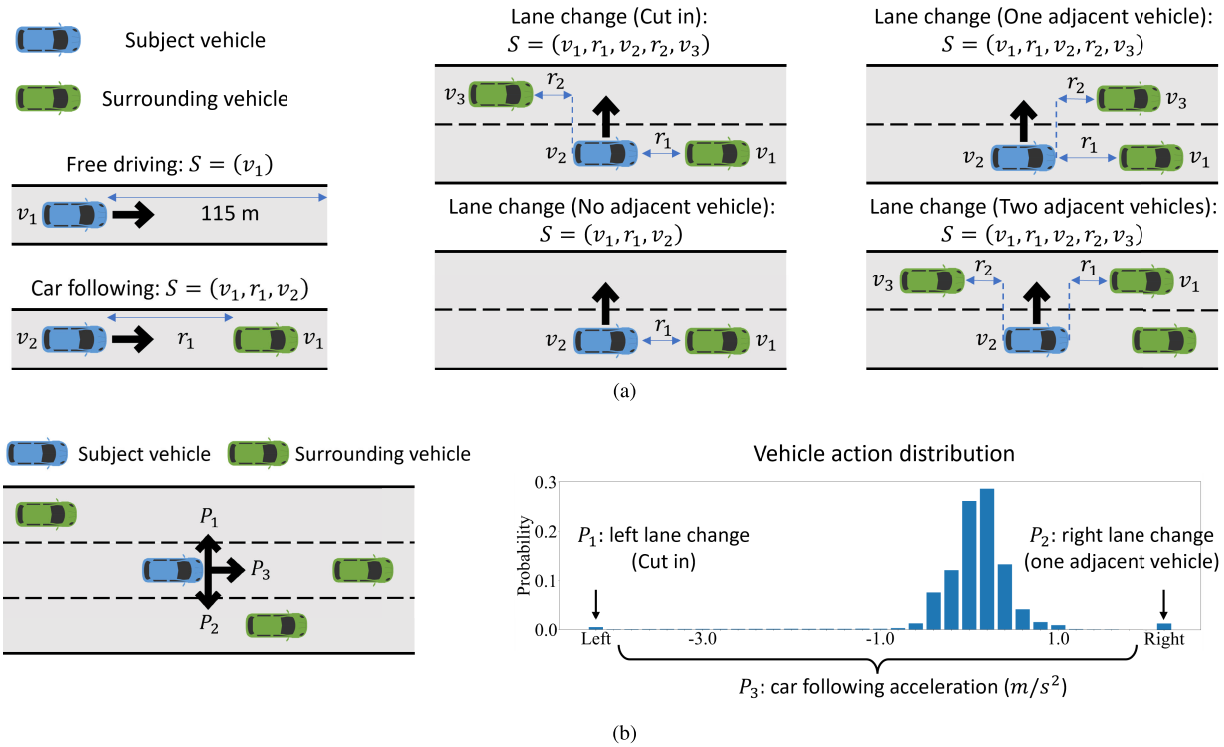


Fig. 3. Illustration of empirical behavior models: (a) longitudinal and lateral behavior models in this study, (b) illustration example of how to simulate a vehicle's action at a decision moment.

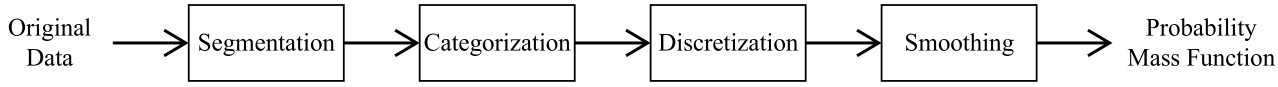


Fig. 4. Data processing flow chart.

$\mathcal{A}$ denotes the action space, and $\mathcal{S}$ denotes the state space. Let $F(S) = \left[P(a_1|S), \cdots F\left(a_{|\mathcal{A}|}|S\right)\right]$ denote the probability mass function under a certain state S, then the empirical behavior model can be denoted by

$$F = \left[F(S_1), \cdots F\left(S_{|\mathcal{S}|}\right)\right] \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{A}|}. \tag{3}$$

Next, we will introduce how to process the NDD and construct F for all six behavior models.

B. Naturalistic Driving Dataset Processing

To construct empirical behavior models, we utilized large-scale NDD from the Integrated Vehicle Based Safety System (IVBSS) dataset [63] and the Safety Pilot Model Deployment (SPMD) dataset [64] at the University of Michigan Transportation Research Institute (UMTRI). In the IVBSS program, 108 drivers ranging from 20 to 70 years old were recruited. Each participant drove the IVBSS vehicle equipped with the data acquisition system (DAS) for 6 weeks. The relative distance and speed with the leading vehicle are recorded by radar at 10 Hz. The SPMD program covered over 34.9 million travel miles and included 98 vehicles equipped with the DAS and Mobileye to record human naturalistic driving behaviors. The data were also recorded at 10 Hz with positions, speeds, and accelerations of ego-vehicles, relative speeds with surrounding vehicles, and both longitudinal and lateral distances between vehicles and lane markings. We queried partial datasets with the following criteria: (1) vehicle was traveling at a speed between 20 m/s and 40 m/s; (2) dry surface condition; (3) daylight condition. The resulting dataset includes approximately 8,200 driving hours data.

The data processing consists of four steps including segmentation, categorization, discretization, and smoothing, as shown in Fig. 4. Specifically, the original data were first segmented into trajectories and then categorized into specific groups based on the six driving situations defined in empirical behavior models. Then, a smoothing technique was applied to the discretized action distribution and finally, we could obtain the probability mass functions for each group, which constituted the six empirical behavior models. More details of the data processing steps can be found in the Appendix. I.

Fig. 5 demonstrates examples of constructed empirical behavior models. Specifically, Fig. 5a and Fig. 5b show examples of the vehicle longitudinal acceleration distributions in free-driving and car-following situations, respectively. For the car-following case, Fig. 5b indicates acceleration distribution when the ego vehicle and its preceding vehicle have the same speed and their range is 30 meters. We can find that for both free-driving and car-following cases, the mean of acceleration is around zero, which is consistent with the intuition. Compared with the car-following situation, the probability of acceleration greater than zero is generally higher in the

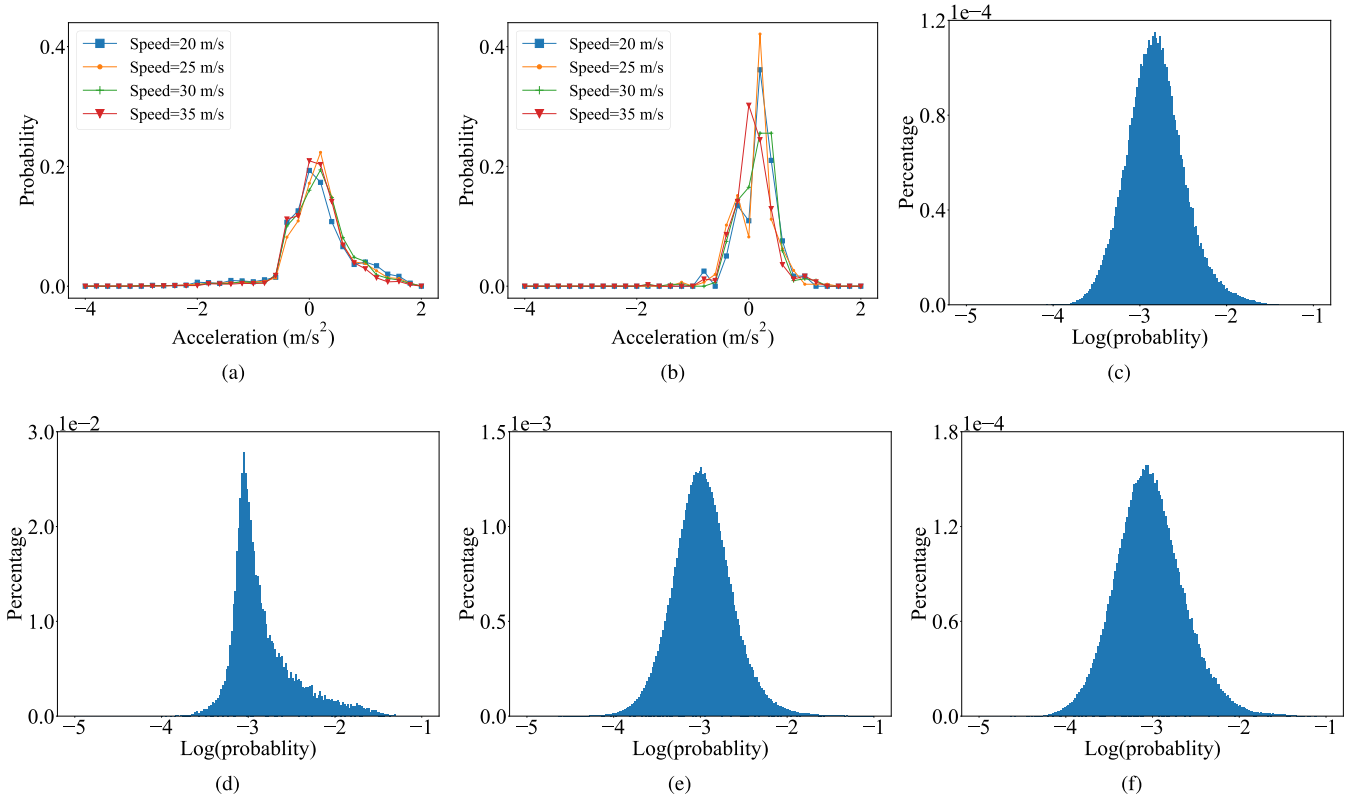


Fig. 5. Examples of empirical behavior models: (a) Free driving, (b) Car following ($r_1 = 30\ m, v_1 = v_2$), (c) Lane change (cut in), (d) Lane change (no adjacent vehicle), (e) Lane change (one adjacent vehicle), (f) Lane change (two adjacent vehicles).

free-driving case, which is reasonable as well. Fig. 5c through Fig. 5f present the lane-change probability distributions across all states within each category, respectively. The x-axis represents the lane-change probability on a logarithmic scale, while the y-axis shows the corresponding percentage. The results indicate that the mean lane-change probability lies around 10^{-3}. This suggests that, on average, a discretionary lane-change maneuver is performed once every two minutes when the decision resolution is 10 Hz, assuming all states occur uniformly. This corresponds to approximately 3.6 kilometers per lane change at a speed of 30 m/s, which is consistent with previous findings that human drivers execute a lane change every 4.5 kilometers on the highway [65].

C. Performance Evaluation of Empirical Behavior Models

In this subsection, the performance of the NDE constructed by the six empirical behavior models is evaluated in a three-lane highway simulation, as illustrated in Fig. 3b (left). The detailed simulation settings can be found in Appendix. II. The Hellinger distance [66] is used to quantitatively measure the dissimilarity between the simulated distribution and the true distribution. The Hellinger distance ranges from 0 to 1, and the smaller the measurement, the better the model performance. To demonstrate the performance, the background vehicles velocity and range distributions, which are important for the AV testing, are investigated as shown in Fig. 6. Results show that although the distributions can roughly capture the trends of the real-world distributions, there still exist significant distributional inconsistency, particularly for the vehicle velocity. This inconsistency is caused by the error accumulation of the empirical behavior models, where the small model errors caused by the limited data quality and quantity are accumulated and amplified along with the simulation steps, as illustrated in Fig. 1. To address this issue, the robust NDE modeling step is developed in the next section to further improve the empirical models.

IV. Robust NDE Modeling

In this section, the robust NDE modeling step is proposed to refine empirical behavior models to minimize the accumulated errors using optimization methods. To achieve this goal, the key is to model the long-term effects of the error accumulation. Specifically, by modeling the vehicle state evolution as a Markov chain, the long-term effects of the error accumulation can be characterized by the stationary distribution of the Markov chain. Then, an optimization problem is formulated to minimize the accumulated errors by adjusting the empirical behavior models, which results in the NDE model with more accurate distributions. In the following paragraphs, we first propose the optimization framework in Section IV-A and then apply the framework on the longitudinal behavior models in Section IV-B.

A. Optimization Framework

In order to solve the error accumulation problem, we need to measure the accumulated error of the NDE generated by the

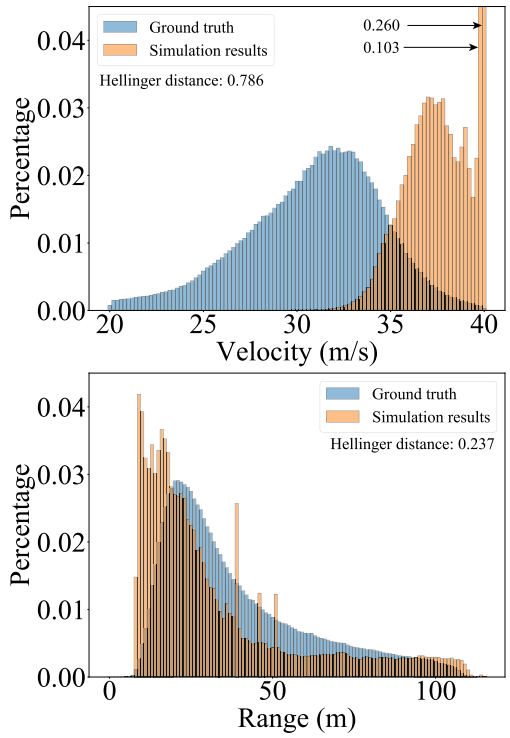


Fig. 6. Velocity and range distributions of the NDE using empirical behavior models.

empirical behavior models. One possible way is to simulate the NDE, collect the data, and obtain the simulated NDE distribution. However, the computational burden of this method is very heavy since a large number of simulations are needed to obtain an accurate estimation. To overcome this problem, we propose to measure the NDE distribution by analyzing the stationary distribution of the NDE Markov chain. By using this analytical method, the NDE stationary distribution serves as an accurate approximation of the simulated environment, which reflects the performance of the empirical behavior models. By fitting the simulated stationary distribution with real-world ground-truth, we can improve the NDE accuracy and solve the error accumulation issue.

Following this idea, the optimization framework can be formulated as follows:

$$\begin{aligned} \min_{F} \quad & \text{distance}\left(F, F^{\star}\right) \\ \text{s.t.} \quad & \text{1. Definition of the stationary distribution.} \\ & \text{2. Relation between behavior model and Markov chain transition probability.} \\ & \text{3. Relation between stationary distribution and ground-truth distribution.} \\ & \text{4. Other constraints.} \end{aligned} \tag{4}$$

Eq. (4) shows the overall formulation of the proposed robust modeling step. The decision variable is the vehicle behavior model F as shown in Eq. (3), and the objective is to minimize the adjustment to the empirical behavior model $F^{\star}$ while ensuring the accuracy of the stationary distribution. To achieve this objective, there are generally four sets of constraints in the optimization formulation. The first set of constraints is the standard definition of stationary distribution, which indicates that the state will always follow its stationary distribution after reaching the steady state. The second set of constraints describes the relationship between the behavior model and Markov chain state transition probability. The stochastic vehicle behavior model outputs actions for the next time step and therefore determines the state transition process. The third set of constraints is to match the stationary distribution of the simulation with the real-world ground-truth distribution, which is the key to reducing the accumulated errors. As a result, the simulated environment can be guaranteed to fit the desired real-world statistics (e.g., velocity and range distributions) even after a long simulation time horizon. The last set of constraints denotes other standard requirements, such as non-negative constraints of probability mass functions, normalization of stationary distribution, etc.

B. Optimization of Longitudinal Behavior Models

In this section, we apply the proposed framework to optimize the two empirical longitudinal behavior models as a proof of concept, while keeping the four empirical lateral behavior models unchanged. As the velocities cannot be well modeled by the empirical models as shown in Fig. 6, we choose the velocity distribution as the optimization target.

For the free-driving behavior model, we define the discretized speed as the state of the Markov chain. It is easy to find that the Markov chain is finite, irreducible, and aperiodic, so there exists a unique positive stationary distribution π [67] satisfying

$$\pi^{T}\mathbf{P} = \pi^{T}, \tag{5}$$

$$\sum_{S\in\mathcal{S}} \pi_S = 1, \tag{6}$$

$$\pi \succcurlyeq 0, \tag{7}$$

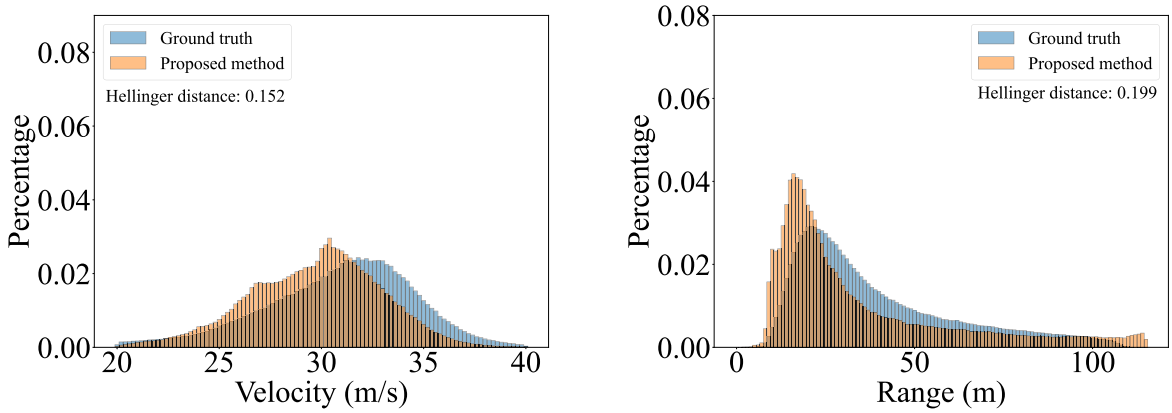
where $\mathbf{P}$ is the state transition probability matrix. As the vehicle longitudinal acceleration depends only on its current speed in the free-driving situation, the state transition probability matrix $\mathbf{P}$ is essentially a function of the behavior model F in Eq. (3) as

$$\mathbf{P}\left(S_i, S_j\right) = G\left(F\right), \forall S_i, S_j \in \mathcal{S}, \tag{8}$$

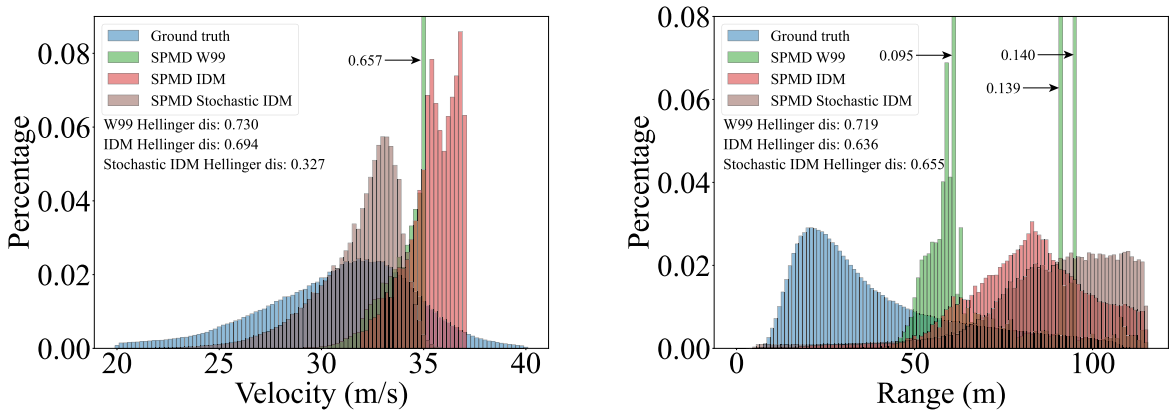
where $G(\cdot)$ is a linear mapping from the longitudinal acceleration to the state transition. For example, if the current speed falls in the state S_i, the next speed after the transition is S_j, the time resolution is Δt, and the probability of taking acceleration a that satisfies the $S_j = S_i + a\cdot\Delta t$ is $p\left(a|S_i\right)$, then $\mathbf{P}\left(S_i, S_j\right) = p\left(a|S_i\right)$. Moreover, as the goal of the optimization is to match the vehicle stationary speed distribution with the real-world speed distribution in the free-driving situation, we have

$$\pi = \pi^{\star}, \tag{9}$$

where $\pi^{\star}$ is the ground-truth of the speed distribution in free-driving situations that is obtained from the large-scale NDD.



(a)



(b)

Fig. 7. Velocity and range distributions of the: (a) proposed method; (b) SUMO simulator.

Finally, the optimization problem can be formulated as below:

$$\min_{F} \quad \left\| F - F^{\star} \right\|_{Frob} \tag{10}$$

$$\text{s.t.} \quad \pi^{T}\mathbf{P} = \pi^{T}, \tag{11}$$

$$G\left(F\right) = \mathbf{P}\left(S_i, S_j\right), \quad \forall S_i, S_j \in \mathcal{S}, \tag{12}$$

$$\pi = \pi^{\star}, \tag{13}$$

$$\sum_{a \in \mathcal{A}} F\left(a|S\right) = 1, \quad \forall S \in \mathcal{S}, \tag{14}$$

$$\sum_{S_j \in \mathcal{S}} \mathbf{P}\left(S_i, S_j\right) = 1, \quad \forall S_i \in \mathcal{S}, \tag{15}$$

$$\sum_{S \in \mathcal{S}} \pi_S = 1, \tag{16}$$

$$F, \mathbf{P}, \pi \succcurlyeq 0. \tag{17}$$

The Frobenius norm $\|\cdot\|_{Frob}$ is adopted to measure the distance between the optimized free-driving model F and empirical free-driving model $F^{\star}$. Comparing with the constraints discussed in Eq. (4), Eq. (11) denotes the definition of the stationary distribution, Eq. (12) denotes the relation between behavior model and transition probability, Eq. (13) denotes the distributional consistency between the stationary distribution and the ground-truth, and Eqs. (14-17) denote other constraints including the normalization requirements for the acceleration probability mass function, state transition probability matrix, and stationary distribution, respectively, and the non-negative requirements. It can be found out that this is a linear programming problem that can be solved efficiently using commercial solvers, for example, Gurobi [68].

For the car-following situation, the vehicle state is composed of the speed of the subject vehicle (v), range (r), and range rate (rr) with the preceding vehicle. The state transition in the car-following situation depends not only on the subject vehicle action but also preceding vehicle action, which makes the optimization problem more complex. To solve this issue, we optimize the steady-state situation of the car-following model, which is a necessary condition regardless of the evolving process to the stationary distribution. As the preceding vehicle has reached the steady-state, the ego-vehicle state transition relies only upon its own action. Then, the optimization problem can be formulated as the same as Eqs. (10-17), where decision variables are the probability mass functions of the car-following accelerations (F in Eq. (3)), and Eq. (11) is a three-dimensional joint state distribution (i.e., v, r, and rr). It is also a linear programming problem that can be solved efficiently.

V. Performance Evaluation

In this section, the performance of the proposed NDE modeling framework is evaluated.

TABLE I
DESCRIPTIVE STATISTICS FOR DIFFERENT METHODS RESULTS

Method	Velocity (m/s)					Range (m)				
	Min	Max	Mean	Range	SD	Min	Max	Mean	Range	SD
Empirical behavior models	23.60	40.00	38.12	16.40	1.83	4.85	114.91	32.72	110.06	24.83
SUMO (SPMD W99)	25.19	35.00	34.53	9.81	0.89	7.04	115.00	73.70	107.96	18.13
SUMO (SPMD IDM)	31.49	37.00	35.45	5.51	1.17	5.10	115.00	82.06	109.90	17.25
SUMO (SPMD Stochastic IDM)	20.83	36.52	31.88	15.69	1.99	5.11	115.00	88.88	109.89	20.00
Proposed method	20.00	40.00	29.72	20.00	3.21	0.004	115.00	34.76	115.00	25.96
Ground truth	20.00	40.00	30.86	20.00	3.62	3.94	114.88	38.89	110.94	22.48

A. Distributional Consistency

As the distributional consistency is critical for the generated NDE, we first evaluate whether the proposed NDE can generate accurate velocity and range distributions, compared with the existing NDE baseline (i.e., SUMO [69]) and the empirical models constructed in Section III. Specifically, three existing car-following models, the Wiedemann 99 model (W99) [70], IDM [23] model, and the stochastic IDM model are selected as car-following model baselines, which are widely applied in existing traffic simulators. The SUMO LC2013 lane-changing model is used for lateral behavior. For fair comparisons, the model parameters are calibrated with the SPMD NDD described in Section III-B using the calibration method developed in [71], and [72]. More simulation settings can be found in Appendix. II.

Fig. 7 shows the results of the proposed NDE model and existing NDE models. It can be found that the proposed NDE model can significantly better reproduce the real-world velocity and range distributions than existing ones. Specifically, both the IDM and Wiedemann 99 models are concentrated in a small interval of velocity and range, while the real-world distributions range among a much wider interval. The stochastic IDM model can generate better velocity results, however, the range distribution is still inaccurate. It is reasonable as these existing models are designed for accident-free purposes and therefore might be more conservative and have a large car-following distance.

Compared with the results before the optimization as shown in Fig. 6, the accuracy of the simulated velocity distribution is also significantly improved, which validates the effectiveness of the robust modeling step. The descriptive statistics, including the mean, range, and standard deviation (SD) are shown in Table. I. From the results, we can find that the proposed method achieves the overall best performance, generating accurate results that align closely with the ground truth data.

To further quantify the performance in terms of the entire distribution, the Hellinger distances of all these models are calculated as listed in Table. II. Hellinger distance is a type of f-divergence commonly used in probability and statistics domains, to measure the similarity between two distributions. With a range of values from 0 to 1, a smaller Hellinger distance signifies a better goodness-of-fit. Based on the results, we can observe that the proposed method outperforms existing SUMO-based methods. Moreover, it shows a significant improvement over the empirical behavior models, which demonstrates the effectiveness of the optimization step.

TABLE II
QUANTITATIVE PERFORMANCE EVALUATION FOR DIFFERENT METHODS

Method	Velocity	Range
Empirical behavior models	0.786	0.237
SUMO (SPMD W99)	0.730	0.719
SUMO (SPMD IDM)	0.694	0.636
SUMO (SPMD Stochastic IDM)	0.327	0.655
Proposed method	**0.152**	**0.199**

In addition to analyzing velocity and range distributions, we also analyzed the frequency of discretionary lane changes by computing the average travel distance per lane change, defined as the total distance traveled by all vehicles divided by the total number of lane changes. Our model yields an average of 4.86 kilometers per discretionary lane change, compared to 0.54 kilometers for SUMO (SPMD Stochastic IDM). In real-world highway conditions, the corresponding value is approximately 4.45 kilometers [65]. These results demonstrate that our method more accurately replicates real-world lane-changing behavior compared to the baseline.

B. AV Testing Using the Proposed NDE

To further demonstrate the capability of the proposed NDE model for AV testing, we test the safety performance of an AV model utilizing the proposed NDE and SUMO (SPMD IDM) environments, respectively. Following previous studies on AV testing [2], [7], [8], [13], the AV accident rate in the NDE is chosen as the measurement, and the Monte Carlo method is applied to estimate the accident rate as shwon in Eq. (1). Specifically, the IDM and MOBIL models [13] are used as the AV model, one simulation test is conducted for a constant driving distance (400 m) of the AV, and the testing result (accident or not) of each simulation is utilized to calculate the accident rate. To obtain accurate evaluation results, the AV will run 5×10^6 simulations in the proposed NDE and SUMO environments, respectively, which is approximately 1.2 million driving miles for the AV under test. The experiments take around four hours using the University of Michigan's Great Lakes High-Performance Computing cluster with 1,300 cores (Intel Xeon Gold 6154 processor) and 9,000 GB RAM.

During the simulation tests in the SUMO environment, there is no accident occurred so the estimated AV accident rate is zero. This is because the SUMO environment is designed for accident-free simulations, and therefore, it cannot effectively evaluate the AV safety performance. During the simulation

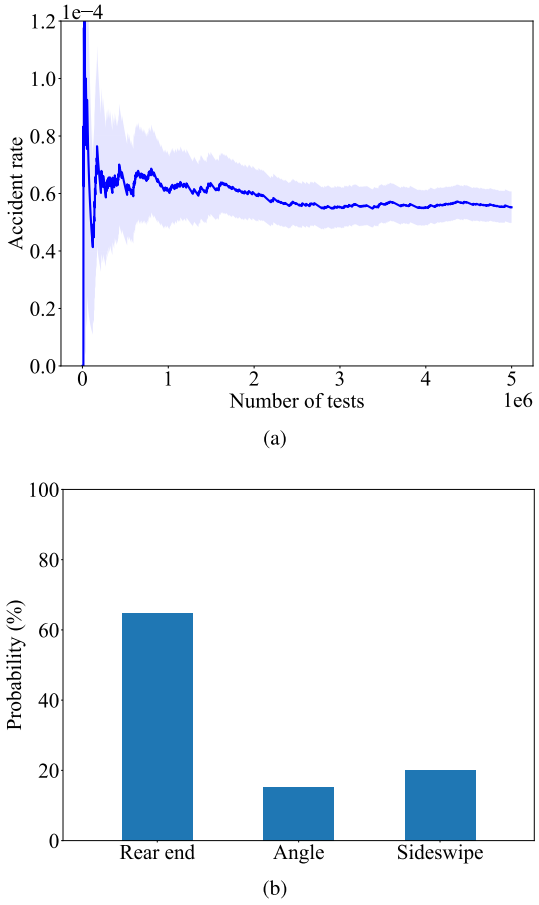


Fig. 8. AV testing and evaluation results using the proposed NDE: (a) estimation results of the accident rate (shaded area denotes the 90% confidence interval), (b) accident type distribution.

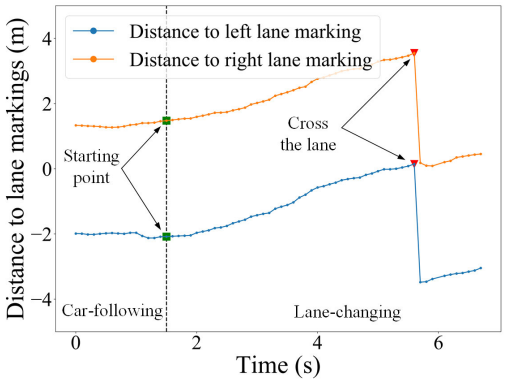


Fig. 9. Lane-changing events identification example: the subject vehicle is doing a left lane change.

tests in the proposed NDE, however, the estimated accident rate of the AV is 5.5×10^{-5} accidents per simulation, where 276 accidents are generated in the AV testing process. The estimation results with the number of tests can be found in Fig. 8a, where the shaded area denotes the 90% confidence interval, and the accident type distribution is shown in Fig. 8b based on the definitions of the National Highway Traffic Safety Administration [73]. The results show that the proposed NDE can successfully generate diverse safety-critical situations to evaluate the AV safety performance. With more distributionally accurate NDE models, the safety performance of AVs could be evaluated more effectively.

VI. Conclusion

In this paper, we propose a distributionally consistent NDE modeling framework for AV testing purposes. We first propose to directly generate empirical behavior models using large-scale NDD. These empirical behavior models can serve as basic models since they will converge in probability to the real behavior models when the data size and quality are sufficiently large and accurate. To address the error accumulation issue and guarantee the accuracy of the NDE throughout the simulation, a robust modeling step based on the Markov process is proposed, which optimizes the empirical models by matching simulated stationary distributions with the ground truth. The proposed method is validated for the multilane highway driving environment using large-scale real-world NDD. The vehicle speed and range distributions generated by the proposed NDE are consistent with the empirical ground truth, which are important for AV testing. Moreover, the generated NDE is utilized to test the safety performance of an AV agent, which further validates the effectiveness of the proposed NDE.

There are certain limitations in the current work that can be addressed by future research. For example, homogeneous vehicles are considered in this study where all background vehicles share the same behavior models and vehicle classes. In the real-world driving environment, there are different vehicle classes, such as sedan, truck, heavy vehicle, etc. Moreover, within a specific vehicle class, they may have different behavior models, for example, some drivers are more aggressive, and some drivers are more conservative. Also, human drivers might behave differently when interacting with an AV, therefore, further work as well as new data is needed to consider the AV influences on surrounding vehicles. In addition, although the proposed method outperforms the existing simulation models, there are still non-negligible differences between the simulated distribution and the real data. As such, additional research efforts are required to further refine the modeling accuracy. We leave these for future research.

Appendix I
Naturalistic Driving Data Processing

The real-world naturalistic driving data often contain discontinuities due to various factors such as data loss or inconsistencies in the detection and tracking of surrounding objects. These factors necessitate a robust preprocessing step to ensure the integrity and reliability of the data used for further analysis. To address these challenges, the first step is to segment the original data into usable trajectories with the following criteria:

- Time continuity: We required that the data should be time-continuous, with no discontinuities exceeding 2 seconds. This choice is based on the observation that the average lane-change duration is approximately 5 seconds [74]. By setting the threshold at 2 seconds, we aimed to minimize the risk of missing the identification of lane-change events in subsequent analyses.

- Consistency in detection: The identification of surrounding vehicles needed to be consistent throughout the trajectory, Frequent changes in detection could lead to incorrect associations and misinterpretation of surrounding agents. Therefore, we require there be no frequent ID switch of agents, and the vehicle speed is between our analysis interval, i.e., 20 to 40 m/s.
- Minimum trajectory length: We set a minimum trajectory length of 3 seconds to ensure that each segment contained enough data points for meaningful analysis.

Second, considering the driving environment of the subject vehicle, trajectory data points will be categorized into the six situations as illustrated in Fig. 3a, correspondingly. An important step in the categorization is to identify the lane-changing event by analyzing the lateral distance to the lane markings. Fig. 9 shows an example that the ego-vehicle is doing a left lane change. Note that the sign of the distance to lane marking differentiates the left and right lane markings. Fig. 9 shows the whole process of a left lane-changing event: the vehicle is approaching the left lane marking and cross the lane when the distance to the left lane marking equals zero, and then change to the maximum (lane width) after entering the target lane. The starting moment and crossing the lane moment of lane-changing events can be identified based on both the distance and slope change of the distance, accordingly. Similar techniques are also utilized for lane-changing detection in a recent study [75]. The starting point of the lane-changing event will be used for calculating the lane change probability.

Third, for each data point, the state was discretized using the following resolutions: except for the speed resolution of longitudinal behavior is 0.2 m/s, the speed and spacing resolutions are 1 m/s and 1 m for all other situations. The longitudinal acceleration discretization resolution is set as 0.2 m/s^2. Therefore, at each decision moment, a vehicle has 33 possible actions: left lane change, 31 longitudinal accelerations (range from -4 to 2 m/s^2 with 0.2 as the discretization resolution), right lane change.

Finally, the Moving Average (MA) [76] smoothing technique was applied. During the smoothing process, states leading to an inevitable crash will not be included. After that, the empirical probability of each action at each state can be calculated by its frequency in the dataset of the corresponding category. These constitute empirical data-driven vehicle behavior models.

Appendix II
Experiment Settings

The experiment settings of the proposed method and the SUMO baseline is discussed in the section. For the proposed method, a three-lane highway driving environment is developed based on an open-source highway traffic simulator [77]. The bicycle model is implemented to update vehicle states at a 10Hz frequency. All lane-changing maneuvers are set completed in 1 second. To be consistent with the NDD, the speed of the NDE simulation is bounded between 20 to 40 m/s, and the longitudinal acceleration is bounded between -4 to 2 m/s^2. Since both the longitudinal and lateral models are data-driven, there are inevitably states where no NDD is collected. We will use the IDM and MOBIL model for states not covered by the proposed behavior models. The IDM model parameters are calibrated from the same NDD. The MOBIL model parameters are partially from [78] which are also calibrated using the SPMD dataset. The detailed model parameters are listed in the next paragraph. The simulation initialization method determines the initial state of all vehicles, which includes the position and speed, etc. A realistic initialization is preferred to shorten the required warm-up time, which can improve the simulation efficiency. In this study, a data-driven initialization method is proposed based on NDD and the details can be found in Appendix. III. Using the proposed initialization method, around 60 vehicles will be initialized for each simulation. It is approximately 1360 vehicles/hour/lane, which belongs to level of service (LOS) C for multilane highways [79]. We ran 100 simulations to mitigate the randomness effect. To fully examine the error accumulation issue, each simulation ran 15 minutes, which includes 10 minutes of warm-up time and 5 minutes of data collection.

The calibrated parameters of the W99 model are: minGap (2.0 m), CC1 (1.2 s), CC2 (9.9 m), CC3 (-19.3 s), CC4 (-0.2 m/s), CC5 (0.1 m/s), CC6 (1.5 $10^{-4} rad/s$), CC7 (1.0 m/s^2), CC8 (0.6 m/s^2), CC9 (0.5m/s^2). The calibrated parameters of the IDM model using the SPMD are: maximum acceleration (0.8 m/s^2), desired speed (37 m/s), exponent parameter (3.0), comfortable deceleration (-1.3 m/s^2), gap at standstill (0.1 m), and desired time headway (0.8 s). To generate a stochastic version IDM, we added zero-mean Gaussian noise to the standard IDM output based on existing literature [40], [42]. We examined different standard deviation parameters from 0.5 to 3.0 with a resolution of 0.5. We presented the results with the best performance (standard deviation equals 1.5). The parameters of the MOBIL model are: politeness factor (0.1), utility threshold (0.2 m/s^2), and maximum safe deceleration (-3 m/s^2). For the SUMO baseline, all simulation settings are the same as the proposed method. Each simulation lasts for 15 minutes, which includes 10 minutes of warm-up time and 5 minutes of data collection. We set the input traffic flow as 1360 vehicles/hour/lane and 60 vehicles will be generated in each simulation. The initial speed of the vehicle is set as 32 m/s based on NDD. To account for the stochasticity of the simulator, we ran 100 simulations for each model to collect data.

Appendix III
The Proposed Data-Driven Simulation Initialization Method

We propose a data-driven initialization method to sequentially determine the $(i+1)$-th vehicle state (x_i, y_i, v_i), i.e., longitudinal position, lateral position, and velocity, of downstream vehicles based on its upstream i-th vehicles' states. The lateral position y_i is the same as the lateral coordinates of its lane center. The first vehicle of each lane is determined by sampling its longitudinal position inside an initial zone from a uniform distribution and its velocity from the empirical

velocity distribution. This can be expressed as

$$x_i \sim U\left(0, d_0\right), \tag{18}$$

$$v_i \sim \pi_v^{\star}, \tag{19}$$

where $U(\cdot,\cdot)$ denotes the uniform distribution, d_0 is the predetermined initial zone size, v_0 is the speed, and $\pi_v^{\star}$ is the empirical velocity distribution obtained from the NDD. Based on the state of the i-th vehicle, we can determine its downstream $(i+1)$-th vehicle state. An indicator variable follows a Bernoulli distribution $I_i \sim B\left(p_{CF}\right)$ is sampled to determine whether the i-th vehicle is in the car-following $(I_i = 1)$ situation or free-driving $(I_i = 0)$ situation. p_{CF} is the probability a vehicle is in the car-following case, which can be approximated from NDD. Then the joint distribution of range r and range rate rr given the current speed $g(r, rr|v_i)$ is queried from the NDD if $I_i = 1$. Then, the position and velocity of the next (downstream) vehicle can be obtained by

$$x_{i+1} = x_i + r, \tag{20}$$

$$y_{i+1} = y_i, \tag{21}$$

$$v_{i+1} = v_i + rr, \tag{22}$$

where r, rr is the range and range rate sampled from $g(r, rr|v_i)$, and the lateral position of the $(i+1)$-th vehicle is the same as the i-th vehicle since they are in the same lane. If $I_i = 0$, the next vehicle is randomly generated in a new initial zone outside the car-following observation range, and the velocity follows the empirical velocity distribution. It can be represented as

$$x_{i+1} = x_i + d_{obs} + r, \tag{23}$$

$$r \sim U\left(0, d_0\right), \tag{24}$$

$$y_{i+1} = y_i, \tag{25}$$

$$v_{i+1} \sim \pi_v^{\star}, \tag{26}$$

where d_{obs} is the predetermined car-following observation range suggested by the NDD. Vehicles on each lane can be sequentially generated by repeating this process. Note that unrealistic initialization that leads to trivial initial collision will be rejected and resampled. Based on the data, the parameters used in the study are $d_0 = 50m$, $p_{CF} = 0.68$, and $d_{obs} = 115m$.

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